Chapter 5 — Discrete Probability Distributions

Chapter 5 is about using probability **when the answer is a number**.

Instead of only asking:

"What is the probability of an event?"

we now ask:

"What values can X take, and what is the probability of each value?"

This chapter covers:

- Random variables
- Discrete probability distributions
- Mean / expected value
- Variance
- Standard deviation
- Binomial distribution

The chapter objectives include constructing probability distributions, finding mean/variance/expected value, and solving binomial probability problems.

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1. What this chapter is about

Imagine tossing two coins.

The outcomes are:

$$HH,\ HT,\ TH,\ TT$$

But in Chapter 5, we do not only care about the letters.

We define a number:

$$X = \text{number of heads}$$

So:

Outcome	X = number of heads
TT	0
HT	1
TH	1
HH	2

Now we can make a probability table:

X	$P(X)$
0	1/4
1	2/4
2	1/4

This exact coin example appears in the chapter when explaining probability distributions.

Simple idea:

Chapter 5 turns outcomes into numbers.

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2. Variable

Definition

A **variable** is a characteristic that can take different values.

Plain English

It is something that can change.

Tiny examples

Situation	Variable
Students in class	height
Patients in hospital	blood pressure
Tossing coins	number of heads
Phone calls	number of people on hold

Exam clue

If the question talks about a number that changes, like number of heads, number of calls, number of successes, number of people, it is a variable.

Memory hint

Variable = **varies** = changes.

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3. Random Variable

Definition

A **random variable** is a variable whose value is determined by chance.

The chapter defines a random variable as a variable, usually represented by X, that has a numerical value determined by chance for each outcome.

Plain English

You do a random experiment, and the answer becomes a number.

Tiny example

Toss two coins.

Let:

$$X = \text{number of heads}$$

Possible values:

$$0,\ 1,\ 2$$

Because the coin toss is random, X is a random variable.

Exam clue

If the question says:

“Let X be the number of…” “random variable X” “find the probability distribution of X”

then you are working with a random variable.

Memory hint

Random variable = **random experiment gives a number**.

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4. Discrete Random Variable

Definition

A **discrete random variable** has a finite or countable number of values.

The chapter says a discrete random variable has either a finite number of values or a countable number of values.

Plain English

Discrete means you can count the answers.

Usually the answers are whole numbers.

Tiny examples

Random variable	Possible values
Number of heads in 3 coin tosses	0, 1, 2, 3
Number of children in a family	0, 1, 2, 3, …
Number of calls received	0, 1, 2, 3, …
Number of defective items	0, 1, 2, 3, …

Exam clue

Use discrete when the answer is a count:

“number of people” “number of heads” “number of successes” “number of calls” “number of defective products”

Memory hint

Discrete = **counting**.

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5. Continuous Random Variable

Definition

A **continuous random variable** has infinitely many possible values on a measurement scale.

The chapter says continuous random variables are associated with measurements on a continuous scale with no gaps.

Plain English

Continuous means you measure it, not count it.

It can have decimals.

Tiny examples

Variable	Why continuous?
Height	can be 170.1 cm, 170.15 cm
Weight	can be 65.3 kg
Time	can be 10.5 minutes
Temperature	can be 37.2°C

Exam clue

Use continuous when the question uses:

“time” “height” “weight” “temperature” “length” “measurement”

Memory hint

Continuous = **measurement with decimals**.

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6. Discrete vs Continuous

Point	Discrete	Continuous
Type	Counted	Measured
Values	Separate values	Any value in interval
Usually	Whole numbers	Decimals possible
Example	number of students	height of students
Chapter	Chapter 5	Chapter 6

Exam trick

If you can say:

"1, 2, 3, 4..."

it is probably discrete.

If you can say:

"1.2, 1.25, 1.257..."

it is probably continuous.

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7. Probability Distribution

Definition

A **probability distribution** lists all possible values of a random variable and the probability of each value.

The chapter states that a probability distribution consists of the values a random variable can assume and the corresponding probabilities, and it can be shown by table, graph, or formula.

Plain English

It is a table that tells us:

"What can X be?" and "How likely is each value?"

Tiny example

Toss two coins.

Let:

$$X = \text{number of heads}$$

X	$P(X)$
0	1/4
1	2/4
2	1/4

This is a probability distribution.

Exam clue

If the question gives a table with:

$$X$$

and

$$P(X)$$

then it is a probability distribution.

Memory hint

Probability distribution = **value + probability table**.

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8. Requirements for a Probability Distribution

For a table to be a valid probability distribution, two rules must be true.
The chapter gives these two requirements: $\sum P(x) = 1$, and every $P(x)$ must be between 0 and 1.

Rule 1

$$\sum P(X) = 1$$

All probabilities must add to 1.

Rule 2

$$0 \leq P(X) \leq 1$$

No probability can be negative. No probability can be greater than 1.

Tiny example

X	$P(X)$
0	0.2
1	0.3
2	0.5

Check:

$$0.2 + 0.3 + 0.5 = 1$$

All probabilities are between 0 and 1.

So this is valid.

Invalid example

X	$P(X)$
0	0.4
1	0.8

Check:

$$0.4 + 0.8 = 1.2$$

This is not valid.

Exam clue

If the question asks:

"Which value completes the probability distribution?"

then add the known probabilities and subtract from 1.

$$\text{missing probability} = 1 - \text{sum of known probabilities}$$

Memory hint

Probability table must total **one whole pizza**.

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9. Mean of a Discrete Random Variable

Definition

The mean of a discrete random variable is the long-run average value of X.

Formula:

$$\mu = \sum xP(x)$$

The chapter gives the mean formula as $\mu = \sum[x \cdot P(x)]$.

Plain English

Multiply each value by its probability, then add.

It tells you the average result if you repeat the experiment many times.

Tiny example

X	$P(X)$
0	0.25
1	0.50
2	0.25

Find mean.

Step 1: Multiply

$$0(0.25) = 0$$

$$1(0.50) = 0.50$$

$$2(0.25) = 0.50$$

Step 2: Add

$$\mu = 0 + 0.50 + 0.50 = 1$$

Final answer

$$\mu = 1$$

Exam clue

If the question asks:

“mean” “expected value” “average value of random variable”

use:

$$\sum xP(x)$$

Memory hint

Mean = **multiply then add**.

---

10. Expected Value

Definition

Expected value is the same idea as the mean of a probability distribution.

Formula:

$$E(X) = \sum xP(x)$$

The chapter says expected value uses $E(X) = \sum X \cdot P(X)$.

Plain English

Expected value means:

“What do I expect on average in the long run?”

It does not always mean the exact answer you will get once.

Tiny example

A game gives:

Profit X	$P(X)$
10 AED	0.5
-4 AED	0.5

$$E(X) = 10(0.5) + (-4)(0.5)$$

$$= 5 - 2 = 3$$

Expected profit:

$$3 \text{ AED}$$

Exam clue

If the question uses money, profit, loss, payoff, expected gain, expected profit, use expected value.

Memory hint

Expected value = **long-run average result**.

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11. Variance of a Discrete Random Variable

Definition

Variance measures how spread out the values of X are around the mean.

Formula 1:

$$\sigma^2 = \sum (x - \mu)^2 P(x)$$

Shortcut formula:

$$\sigma^2 = \sum x^2 P(x) - \mu^2$$

The chapter gives both variance formulas, including the shortcut formula.

Plain English

Variance tells you how far the values are from the average.

Large variance = values are spread out. Small variance = values are close together.

Exam clue

If the question asks for variance, use:

$$\sigma^2 = \sum x^2 P(x) - \mu^2$$

This shortcut is usually faster.

Memory hint

Variance has the square:

$$\sigma^2$$

So variance = **spread squared**.

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12. Standard Deviation of a Discrete Random Variable

Definition

Standard deviation is the square root of variance.

Formula:

$$\sigma = \sqrt{\sigma^2}$$

The chapter states that the standard deviation of a probability distribution is the square root of the variance.

Plain English

Standard deviation is spread, but in the original units.

Exam clue

If the question asks for standard deviation:

1. Find variance.
2. Square root it.

Memory hint

Standard deviation = **√variance**.

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13. Example: Die Toss Mean and Standard Deviation

The chapter gives this example: find the mean and variance of the number of spots when a die is tossed. The distribution is $X = 1, 2, 3, 4, 5, 6$, each with probability $1/6$.

X	1	2	3	4	5	6
$P(X)$	1/6	1/6	1/6	1/6	1/6	1/6

A) Find the mean

Step 1: Formula

$$\mu = \sum xP(x)$$

Step 2: Substitute

$$\mu = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right)$$

Step 3: Add numerator

$$\mu = \frac{1+2+3+4+5+6}{6}$$

$$\mu = \frac{21}{6}$$

Step 4: Calculate

$$\mu = 3.5$$

Answer

$$\mu = 3.5$$

Plain meaning

If you roll a die many times, the average result will be about 3.5.

B) Find the variance

Step 1: Shortcut formula

$$\sigma^2 = \sum x^2 P(x) - \mu^2$$

Step 2: Find $\sum x^2 P(x)$

$$1^2\left(\frac{1}{6}\right) + 2^2\left(\frac{1}{6}\right) + 3^2\left(\frac{1}{6}\right) + 4^2\left(\frac{1}{6}\right) + 5^2\left(\frac{1}{6}\right) + 6^2\left(\frac{1}{6}\right)$$

$$= \frac{1+4+9+16+25+36}{6}$$

$$= \frac{91}{6}$$

$$= 15.1667$$

Step 3: Subtract μ^2

$$\sigma^2 = 15.1667 - (3.5)^2$$

$$\sigma^2 = 15.1667 - 12.25$$

$$\sigma^2 = 2.9167$$

Answer

$$\sigma^2 \approx 2.917$$

C) Find standard deviation

$$\sigma = \sqrt{2.917}$$

$$\sigma \approx 1.71$$

Answer

$$\sigma \approx 1.71$$

Exam hint

For a probability distribution table:

1. Find $\mu = \sum xP(x)$
2. Find $\sum x^2 P(x)$
3. Use $\sigma^2 = \sum x^2 P(x) - \mu^2$
4. Use $\sigma = \sqrt{\sigma^2}$

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14. Example: People on Hold

The chapter gives this distribution for the number of people placed on hold when they call a radio talk show with four phone lines.

X	0	1	2	3	4
$P(X)$	0.18	0.34	0.23	0.21	0.04

Find the standard deviation.

Step 1: Find the mean

Formula:

$$\mu = \sum xP(x)$$

Substitute:

$$\mu = 0(0.18) + 1(0.34) + 2(0.23) + 3(0.21) + 4(0.04)$$

Calculate:

$$\mu = 0 + 0.34 + 0.46 + 0.63 + 0.16$$

$$\mu = 1.59$$

Step 2: Find $\sum x^2 P(x)$

$$0^2(0.18) + 1^2(0.34) + 2^2(0.23) + 3^2(0.21) + 4^2(0.04)$$

$$= 0 + 0.34 + 4(0.23) + 9(0.21) + 16(0.04)$$

$$= 0 + 0.34 + 0.92 + 1.89 + 0.64$$

$$= 3.79$$

Step 3: Find variance

$$\sigma^2 = \sum x^2 P(x) - \mu^2$$

$$\sigma^2 = 3.79 - (1.59)^2$$

$$\sigma^2 = 3.79 - 2.5281$$

$$\sigma^2 = 1.2619$$

Approximately:

$$\sigma^2 = 1.26$$

Step 4: Find standard deviation

$$\sigma = \sqrt{1.26}$$

$$\sigma \approx 1.12$$

Final answer

$$\sigma \approx 1.12$$

Exam hint

If the question asks standard deviation, do not stop at variance.

You must square root.

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15. Example: Expected Profit

The chapter gives this example: a ski resort loses $70,000 in a bad season and makes $250,000 in a good season. The probability of a good season is 40%. Find expected profit.

Profit X	$P(X)$
250,000	0.40
-70,000	0.60

Step 1: Identify the topic

This is expected value because it asks expected profit.

Step 2: Formula

$$E(X) = \sum xP(x)$$

Step 3: Substitute

$$E(X) = 250000(0.40) + (-70000)(0.60)$$

Step 4: Calculate

$$250000(0.40) = 100000$$

$$-70000(0.60) = -42000$$

Step 5: Add

$$E(X) = 100000 - 42000$$

$$E(X) = 58000$$

Final answer

$$\$58,000$$

Exam hint

Money + probability = expected value.

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16. Binomial Distribution

Definition

A **binomial experiment** is a probability experiment with four conditions:

1. Fixed number of trials.
2. Each trial has only two outcomes: success or failure.
3. Trials are independent.
4. Probability of success stays constant.

The chapter lists these same binomial requirements.

Plain English

Binomial means:

You repeat something a fixed number of times, and each time the answer is yes/no.

Examples:

Situation	Success	Failure
Toss coin 5 times	head	tail
Test 10 products	defective	not defective
Survey 5 people	likes product	does not like product
Patient test	positive	negative

Exam clue

Use binomial when the question has:

“n trials” “sample of 5” “probability is 20%” “success/failure” “exactly x” “at least one” “at most”

Memory hint

Binomial = **bi** = two outcomes.

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17. Binomial Notation

Symbol	Meaning
n	number of trials
x or r	number of successes
p	probability of success
q	probability of failure
$q = 1 - p$	failure probability

The chapter uses p for probability of success, $q = 1 - p$ for failure, n for number of trials, and X for number of successes.

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18. Binomial Probability Formula

Formula

$$P(X = r) = {}_nC_r p^r q^{n-r}$$

or:

$$P(X = x) = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

What it means

Part	Meaning
${}_nC_r$	how many ways successes can happen
p^r	probability of the successes
q^{n-r}	probability of the failures

Plain English

To find exactly r successes:

1. Count how many ways success positions can happen.
2. Multiply by probability of success.
3. Multiply by probability of failure.

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19. Binomial Keywords

Wording	Meaning
exactly 2	$P(X = 2)$
at least 2	$P(X \geq 2)$
at most 2	$P(X \leq 2)$
fewer than 2	$P(X < 2)$
more than 2	$P(X > 2)$
no successes	$P(X = 0)$
at least one	$1 - P(X = 0)$

---

20. Binomial Example: Exactly 2 Successes

Suppose:

$$n = 5, \quad p = 0.20, \quad q = 0.80$$

Find probability exactly 2 people are in favor.

Step 1: Identify values

$$n = 5$$

$$x = 2$$

$$p = 0.20$$

$$q = 0.80$$

Step 2: Formula

$$P(X = 2) = {}_5C_2(0.20)^2(0.80)^3$$

Step 3: Find combination

$${}_5C_2 = \frac{5!}{2!3!}$$

$$= \frac{5 \times 4}{2 \times 1}$$

$$= 10$$

Step 4: Substitute

$$P(X=2) = 10(0.20)^2(0.80)^3$$

Step 5: Calculate powers

$$(0.20)^2 = 0.04$$

$$(0.80)^3 = 0.512$$

Step 6: Multiply

$$P(X=2) = 10(0.04)(0.512)$$

$$= 0.2048$$

Final answer

$$0.2048$$

Exam hint

“Exactly” means use the binomial formula once.

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21. Binomial Example: At Least One

Suppose 20% of people like a new product. A sample of 5 people is selected. Find probability that at least one person likes it.

This is like your final exam question.

Step 1: Identify values

$$n = 5$$

$$p = 0.20$$

$$q = 0.80$$

Step 2: Understand "at least one"

At least one means:

$$1, 2, 3, 4, 5$$

Instead of calculating all of these, use complement:

$$P(X \geq 1) = 1 - P(X = 0)$$

Step 3: Find $P(X = 0)$

No one likes it means all 5 do not like it.

$$P(X = 0) = (0.80)^5$$

$$P(X = 0) = 0.32768$$

Step 4: Subtract from 1

$$P(X \geq 1) = 1 - 0.32768$$

$$= 0.67232$$

Final answer

$$0.67232$$

Exam hint

"At least one" = easiest complement question.

$$1 - P(0)$$

---

22. Mean, Variance, and Standard Deviation for Binomial Distribution

For binomial distribution:

Mean

$$\mu = np$$

Variance

$$\sigma^2 = npq$$

Standard deviation

$$\sigma = \sqrt{npq}$$

The chapter objective includes finding mean, variance, and standard deviation for a binomial distribution.

Tiny example

Given:

$$n = 5, \quad p = 0.20, \quad q = 0.80$$

Find mean and standard deviation.

Step 1: Mean

$$\mu = np$$

$$\mu = 5(0.20)$$

$$\mu = 1$$

Step 2: Variance

$$\sigma^2 = npq$$

$$\sigma^2 = 5(0.20)(0.80)$$

$$\sigma^2 = 0.80$$

Step 3: Standard deviation

$$\sigma = \sqrt{0.80}$$

$$\sigma = 0.894$$

Final answer

$$\mu = 1, \quad \sigma \approx 0.89$$

Exam hint

For binomial:

Mean is not p. Mean is:

$$np$$

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23. Solving Final Exam Questions from Chapter 5

These are the final exam questions connected to Chapter 5.

Question 11

What probability value would be needed to complete the probability distribution?

Visible probabilities:

$$0.09,\ ?,\ 0.15,\ 0.34,\ 0.27$$

Step 1: Identify the topic

This is a probability distribution table.

For any probability distribution:

$$\sum P(X) = 1$$

Step 2: Add the known probabilities

$$0.09 + 0.15 + 0.34 + 0.27$$

$$= 0.85$$

Step 3: Find the missing value

$$1 - 0.85 = 0.15$$

Final answer

$$0.15$$

Choose the option that says:

0.15

Exam clue

Missing probability in a distribution:

$$1 - \text{sum of all other probabilities}$$

Common mistake

Adding the X values instead of adding the probabilities.

Only probabilities must add to 1.

Question 12

The expected value $E(X)$ in the above table is:

From the visible table, we use:

$$X = -5, -3, 0, 1, 6$$

$$P(X) = 0.09, 0.15, 0.15, 0.34, 0.27$$

Step 1: Identify the topic

Expected value.

Use:

$$E(X) = \sum xP(x)$$

Step 2: Multiply each X by its probability

X	$P(X)$	(xPx)
-5	0.09	$-5(0.09) = -0.45$
-3	0.15	$-3(0.15) = -0.45$
0	0.15	$0(0.15) = 0$
1	0.34	$1(0.34) = 0.34$
6	0.27	$6(0.27) = 1.62$

Step 3: Add

$$E(X) = -0.45 - 0.45 + 0 + 0.34 + 1.62$$

$$E(X) = 1.06$$

Final answer

$$E(X) = 1.06$$

If 1.06 is not in the options, choose:

None of these answers

Exam clue

Expected value = multiply each x by its probability, then add.

Common mistake

Doing:

$$\frac{-5 - 3 + 0 + 1 + 6}{5}$$

That is not expected value because it ignores probabilities.

Question 13

The standard deviation of the random variable X in the above table is:

Use:

$$X = -5, -3, 0, 1, 6$$

$$P(X) = 0.09, 0.15, 0.15, 0.34, 0.27$$

We already found:

$$\mu = 1.06$$

Step 1: Formula

Shortcut formula:

$$\sigma^2 = \sum x^2 P(x) - \mu^2$$

Then:

$$\sigma = \sqrt{\sigma^2}$$

Step 2: Calculate $x^2 P(x)$

X	$P(X)$	$x^2 P(x)$
-5	0.09	$25(0.09) = 2.25$
-3	0.15	$9(0.15) = 1.35$
0	0.15	$0(0.15) = 0$
1	0.34	$1(0.34) = 0.34$
6	0.27	$36(0.27) = 9.72$

Step 3: Add

$$\sum x^2 P(x) = 2.25 + 1.35 + 0 + 0.34 + 9.72$$

$$= 13.66$$

Step 4: Variance

$$\sigma^2 = 13.66 - (1.06)^2$$

$$\sigma^2 = 13.66 - 1.1236$$

$$\sigma^2 = 12.5364$$

Step 5: Standard deviation

$$\sigma = \sqrt{12.5364}$$

$$\sigma \approx 3.54$$

Final answer

$$\sigma \approx 3.54$$

If 3.54 is not in the options, choose:

None of these answers

Exam clue

Standard deviation is always:

$$\sqrt{\text{variance}}$$

Common mistake

Giving 12.5364 as the answer. That is variance, not standard deviation.

Question 14

A food company launched a new product and 20% of people liked it. A sample of 5 consumers were randomly selected. Find the probability that at least one person was in favor.

Step 1: Identify the distribution

This is binomial because:

Binomial condition	In this question
Fixed number of trials	5 people
Two outcomes	in favor / not in favor
Constant probability	20% each person
Independent trials	randomly selected

Step 2: Define values

$$n = 5$$

$$p = 0.20$$

$$q = 1 - p = 0.80$$

Step 3: Understand “at least one”

“At least one” means:

$$1, 2, 3, 4, 5$$

Use complement:

$$P(X \geq 1) = 1 - P(X = 0)$$

Step 4: Find $P(X = 0)$

No one is in favor means all 5 are not in favor.

$$P(X = 0) = (0.80)^5$$

$$P(X = 0) = 0.32768$$

Step 5: Subtract from 1

$$P(X \geq 1) = 1 - 0.32768$$

$$P(X \geq 1) = 0.67232$$

Final answer

$$0.67232$$

So the answer is:

B) 0.67232

Exam clue

“At least one” = $1 - P(0)$

Common mistake

Calculating exactly one:

$$_5C_1(0.20)^1(0.80)^4$$

That gives only one person in favor, not at least one.

Question 15

Same food company question: find the mean and standard deviation for the number of people who were in favor.

Given:

$$n = 5$$

$$p = 0.20$$

$$q = 0.80$$

Step 1: Mean formula

$$\mu = np$$

Step 2: Substitute

$$\mu = 5(0.20)$$

$$\mu = 1$$

Step 3: Standard deviation formula

$$\sigma = \sqrt{npq}$$

Step 4: Substitute

$$\sigma = \sqrt{5(0.20)(0.80)}$$

Step 5: Calculate inside the square root

$$5(0.20)(0.80) = 0.80$$

Step 6: Square root

$$\sigma = \sqrt{0.80}$$

$$\sigma \approx 0.89$$

Final answer

$$\mu = 1, \quad \sigma \approx 0.89$$

Choose the option that says:

$$\mu = 1,\ \sigma = 0.89$$

Exam clue

For binomial mean and standard deviation:

$$\mu = np$$

$$\sigma = \sqrt{npq}$$

Common mistake

Using:

$$\sigma = npq$$

That is variance, not standard deviation.

Chapter 5 Mini Cheat Sheet

Topic	Formula / Rule
Probability distribution rule 1	$\sum P(X) = 1$
Probability distribution rule 2	$0 \leq P(X) \leq 1$
Mean / expected value	$\mu = E(X) = \sum xP(x)$
Variance	$\sigma^2 = \sum (x-\mu)^2 P(x)$
Shortcut variance	$\sigma^2 = \sum x^2 P(x) - \mu^2$
Standard deviation	$\sigma = \sqrt{\sigma^2}$
Binomial failure probability	$q = 1 - p$
Binomial probability	$P(X = r) = {}_nC_r p^r q^{n-r}$
Binomial mean	$\mu = np$
Binomial variance	$\sigma^2 = npq$
Binomial standard deviation	$\sigma = \sqrt{npq}$

When You See This Wording, Use This Method

Wording in exam	Use this
“random variable X”	List possible values of X
“probability distribution”	Check $P(X)$ values and total
“complete the distribution”	$1-$ sum of known probabilities
“expected value”	$\sum xP(x)$
“mean of random variable”	$\sum xP(x)$
“variance”	$\sum x^2P(x) - \mu^2$
“standard deviation”	square root of variance
“two outcomes”	binomial may be used
“fixed number of trials”	binomial clue
“exactly x successes”	binomial formula
“at least one”	$1 - P(0)$
“mean of binomial”	np
“standard deviation of binomial”	$\sqrt{npq}$

Most Likely Chapter 5 Exam Question Types

1. Identify discrete vs continuous random variables.
2. Complete a probability distribution table.
3. Check whether a probability distribution is valid.
4. Find expected value / mean.
5. Find variance.
6. Find standard deviation.
7. Recognize a binomial experiment.
8. Solve exactly x successes using binomial formula.
9. Solve “at least one” using complement.
10. Find binomial mean and standard deviation.

Quick Practice With Answers

Practice 1

A probability distribution has probabilities:

$$0.20,\ 0.30,\ ?,\ 0.10$$

Find the missing probability.

Solution

$$0.20 + 0.30 + 0.10 = 0.60$$

$$1 - 0.60 = 0.40$$

Answer:

$$0.40$$

Practice 2

X	0	1	2
$P(X)$	0.2	0.5	0.3

Find expected value.

Solution

$$E(X) = 0(0.2) + 1(0.5) + 2(0.3)$$

$$= 0 + 0.5 + 0.6$$

$$= 1.1$$

Answer:

$$1.1$$

Practice 3

Using the same table, find $\sum x^2 P(x)$.

Solution

$$0^2(0.2) + 1^2(0.5) + 2^2(0.3)$$

$$= 0 + 0.5 + 4(0.3)$$

$$= 0 + 0.5 + 1.2$$

$$= 1.7$$

Answer:

$$1.7$$

Practice 4

Using the same table, find variance.

We know:

$$\mu = 1.1$$

$$\sum x^2 P(x) = 1.7$$

Solution

$$\sigma^2 = 1.7 - (1.1)^2$$

$$= 1.7 - 1.21$$

$$= 0.49$$

Answer:

$$0.49$$

Practice 5

Using the same table, find standard deviation.

Solution

$$\sigma = \sqrt{0.49}$$

$$= 0.7$$

Answer:

$$0.7$$

Practice 6

A quiz has 4 true/false questions. Probability of guessing correctly is 0.5. Find probability of exactly 3 correct.

Step 1: Values

$$n = 4, \quad x = 3, \quad p = 0.5, \quad q = 0.5$$

Step 2: Formula

$$P(X = 3) = {}_4C_3(0.5)^3(0.5)^1$$

Step 3: Combination

$${}_4C_3 = 4$$

Step 4: Calculate

$$P(X = 3) = 4(0.125)(0.5)$$

$$= 4(0.0625)$$

$$= 0.25$$

Answer:

$$0.25$$

Practice 7

A sample of 6 people is selected. Probability a person likes the product is 0.30. Find probability at least one likes it.

Step 1: Values

$$n = 6, \quad p = 0.30, \quad q = 0.70$$

Step 2: Complement

$$P(X \geq 1) = 1 - P(X = 0)$$

$$= 1 - (0.70)^6$$

$$= 1 - 0.117649$$

$$= 0.882351$$

Answer:

$$0.8824$$

Final Chapter 5 Memory Box

Memorize these:

$$\mu = E(X) = \sum xP(x)$$

$$\sigma^2 = \sum x^2 P(x) - \mu^2$$

$$\sigma = \sqrt{\sigma^2}$$

For binomial:

$$q = 1 - p$$

$$P(X = r) = {}_nC_r p^r q^{n-r}$$

$$\mu = np$$

$$\sigma = \sqrt{npq}$$

And the biggest exam shortcut:

“At least one” = $1 - P(0)$.